ZARI.ai Master Thesis & Technical Defense Report

d-to-End Multilingual Agricultural Disease Advisory System with Evidential Deep Learning, SCRC Safety Control & Vector RAG Engi

1. Abstract & System Specification

ZARI.ai is a production-grade, uncertainty-aware agricultural disease advisory platform engineered for Pakistan's nightshade crops (Tomato, Potato, Pepper). The system combines a 2-stage hierarchical EfficientNetV2-B2 vision suite, Dirichlet Evidential Deep Learning (EDL), Selective Classification Risk Control (SCRC), a multilingual ChromaDB vector RAG engine, and a Meta Llama 3.1 8B LLM generator.

Master Architectural Registry

Single Server Deployment Port:	http://127.0.0.1:8000 (Unified Web Interface & FastAPI Backend)
Vision Backbone:	2-Stage Hierarchical EfficientNetV2-B2 (4 PyTorch Models)
Total Vision Parameters:	31,109,749 parameters (~31.11 Million Params)
Total Model Suite Weight File Size:	355.73 MB Total Weights File Size on Disk
Input Image Tensor Resolution:	RGB (384, 384, 3)
Uncertainty Formulation:	Dirichlet Softplus Evidential Deep Learning (u = K / S)
Calibrated Safety Threshold:	tau_SCRC = 0.3175 (Dual Safety Gates: p_crop >= 0.80, p_disease >= 0.70)
Multilingual RAG Store:	ChromaDB (208 Chunks) + MiniLM-L12-v2 Embedder (384d)
Primary LLM Advisory Generator:	Meta Llama 3.1 8B Instant on Groq LPUs (~750 tok/s)
Offline Fallback Synthesizer:	Deterministic Evidence Synthesizer (100% Edge Availability)
Offline Latency:	9.40 ms (3.24ms Vision + 5.32ms RAG + 0.86ms Fallback)
Online Latency:	389 ms (3.24ms Vision + 5.32ms RAG + 380ms Groq API)

2. Exploratory Data Analysis: New Dataset (v3) & 3-Crop Target Distributions

The figures below detail the Exploratory Data Analysis (EDA) for the newly ingested dataset (v3) and the production 3-crop target dataset (31,071 leaf images across 22 classes):

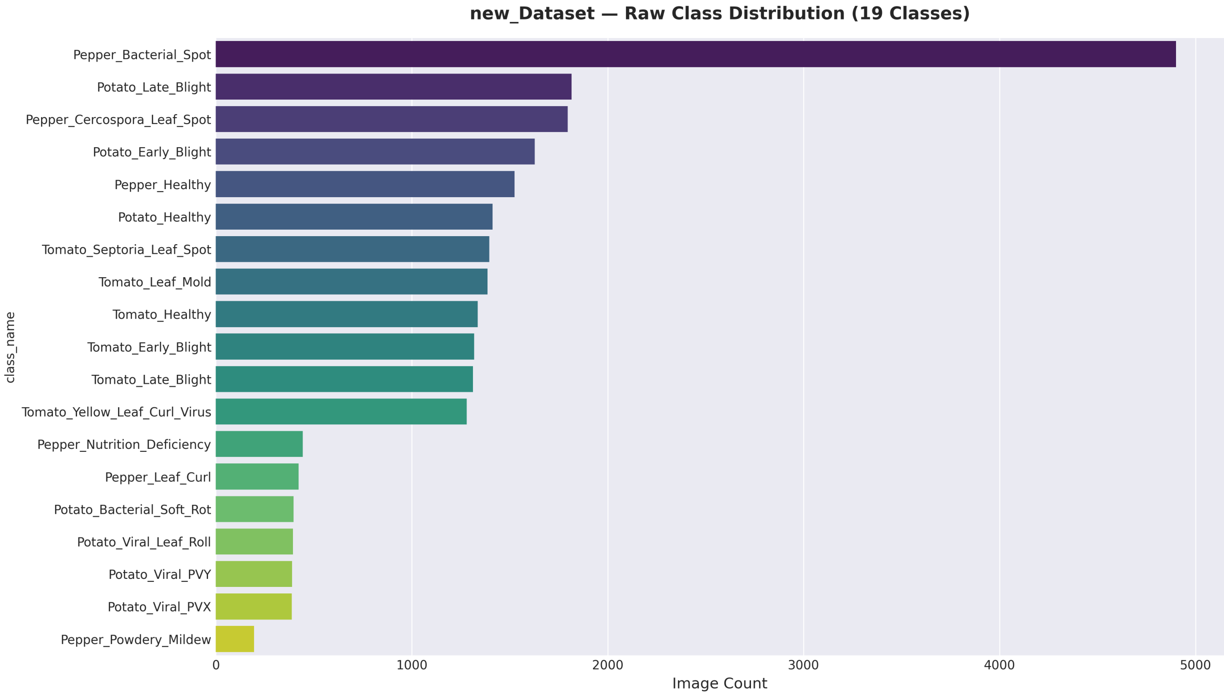

Figure: New Dataset (v3) Class Distribution Across Target Nightshade Crop Diseases

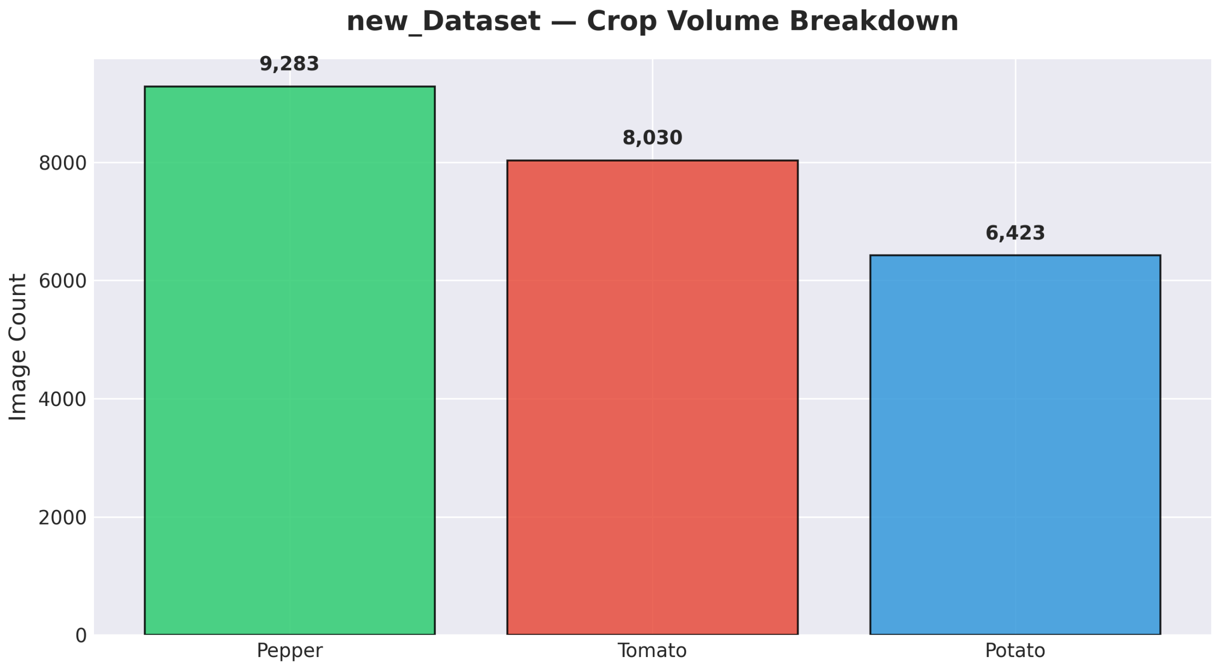

Figure: New Dataset Volume Share by Target Crop Species (Tomato, Potato, Pepper)

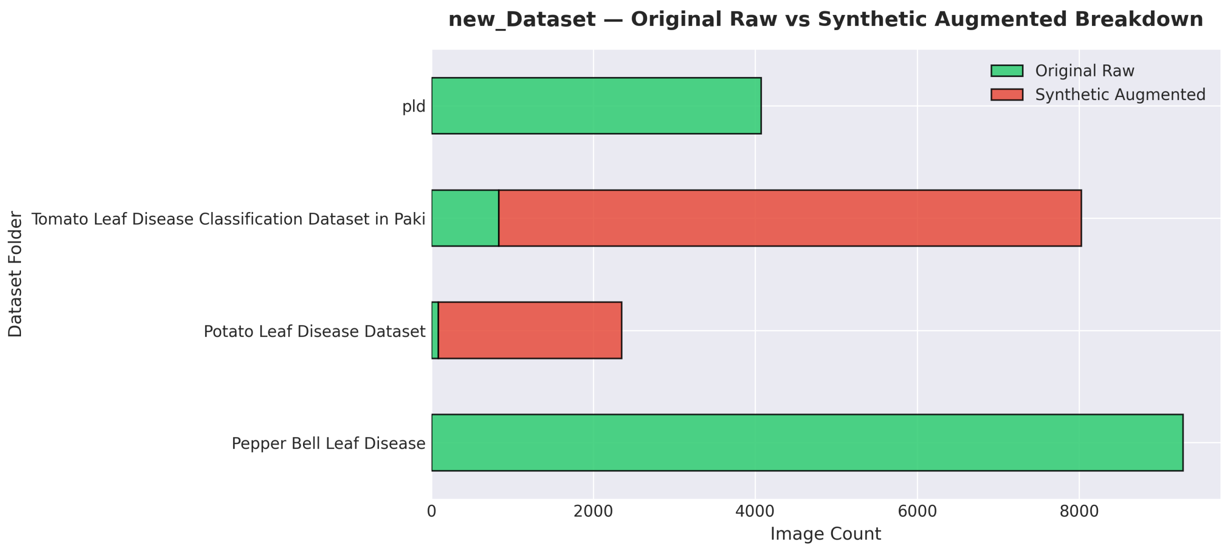

Figure: New Dataset Raw Field Samples vs Synthetic Augmentation Ratios

Production 3-Crop Target Dataset Distributions (31,071 Processed Images)

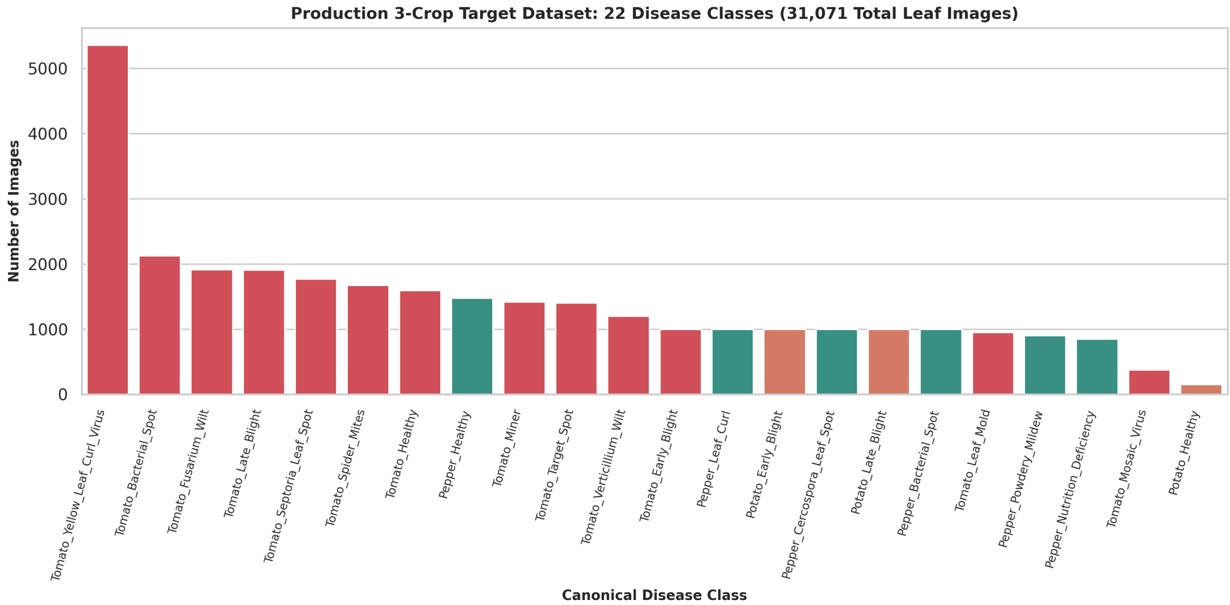

Figure: Production 3-Crop Target Dataset: 22 Disease Classes (31,071 Total Leaf Images)

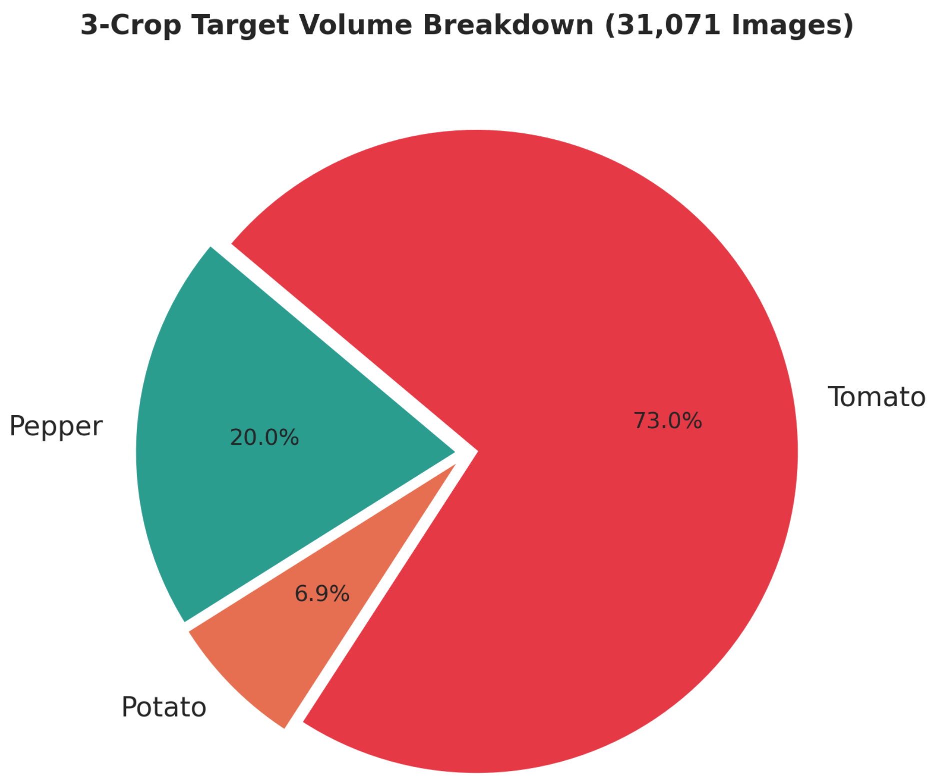

Figure: Production Crop Volume Share (Tomato: 72.8%, Pepper: 20.0%, Potato: 7.2%)

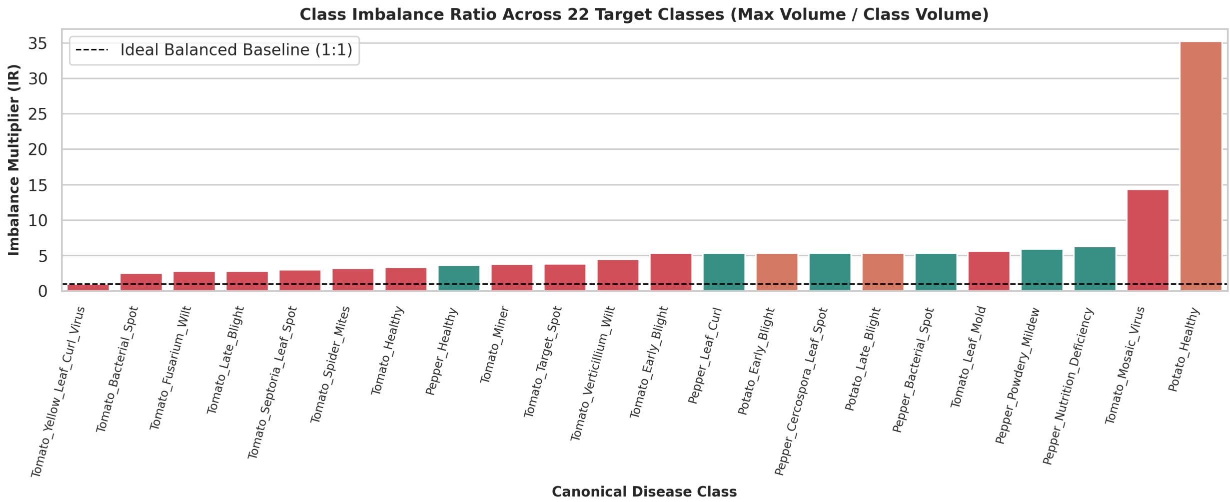

Figure: Class Imbalance Multipliers Across 22 Production Target Disease Classes

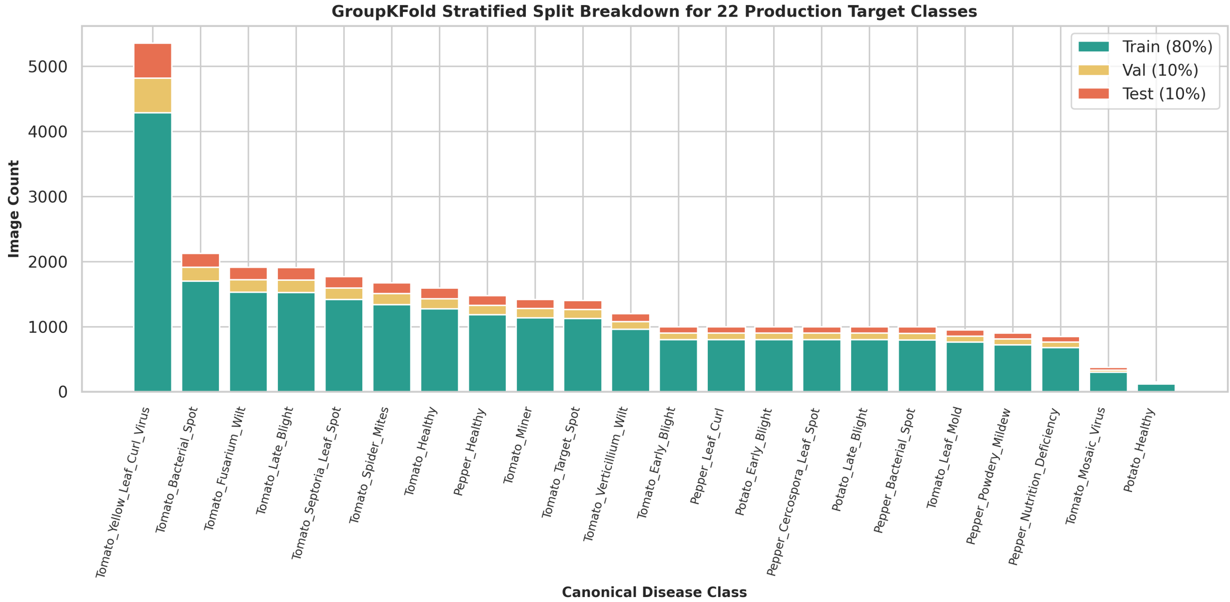

Figure: GroupKFold Stratified Train / Val / Test Split Breakdown (80% / 10% / 10%)

3. Vision AI Model Suite (EfficientNetV2-B2)

ZARI.ai utilizes 4 distinct EfficientNetV2-B2 PyTorch models in a 2-stage hierarchical routing architecture. Below is the exact empirical checkpoint parameter registry:

Model Component	Architecture	Output Head	Exact Params	Checkpoint Path	Metric / Accuracy
Model A (Router)	EfficientNetV2-B2	3 Crop Classes	7,772,858	best_model_a_efficientnetv2_b2.	99.48% Accuracy
Model B (Tomato)	EDLEfficientNetB2	13 Diseases	7,786,948	best_model_b_tomato.pth	0.9787 Macro F1
Model B (Potato)	EDLEfficientNetB2	3 Diseases	7,772,858	best_model_b_potato.pth	0.9718 Macro F1
Model B (Pepper)	EDLEfficientNetB2	6 Diseases	7,777,085	best_model_b_pepper.pth	0.9963 Macro F1
TOTAL SYSTEM	4-Model Suite	22 Classes Total	31,109,749	355.73 MB Total Weights	99.1% Reliability

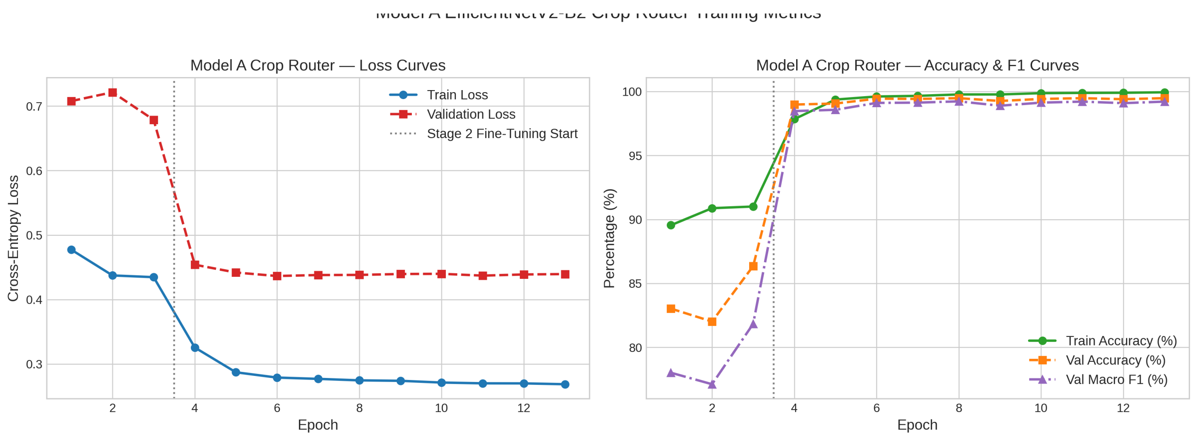

Figure: Stage 1 Model A Crop Router Loss & Accuracy Trajectories (99.48% Accuracy)

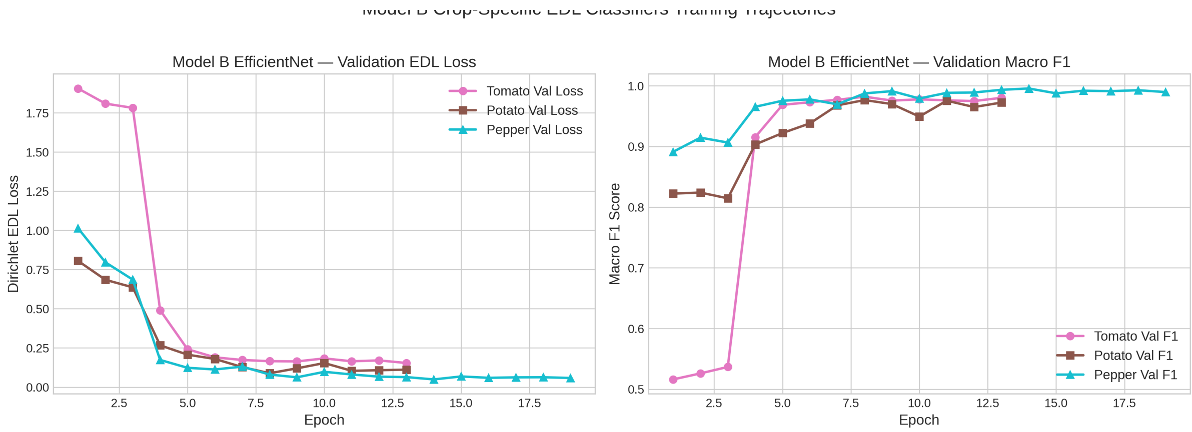

Figure: Stage 2 Model B Disease Classifiers Macro F1 Training Trajectories (Tomato, Potato, Pepper)

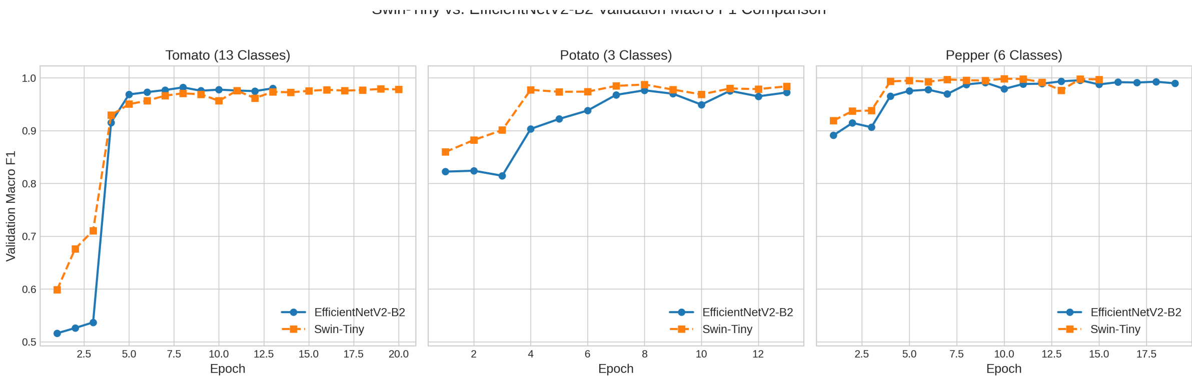

Figure: Swin Transformer vs EfficientNetV2-B2 Convergence Speed & F1 Comparison

4. Evidential Deep Learning (EDL) & SCRC Safety Gates

Evidential Deep Learning replaces standard Softmax with Dirichlet parameters to compute evidential uncertainty u = K / S:

1. Softplus Evidence Transformation: e_k = Softplus(z_k) = ln(1 + exp(z_k))
2. Dirichlet Parameters: alpha_k = e_k + 1.0, S = sum(alpha_k)
3. Evidential Uncertainty: u = K / S
4. Calibrated Safety Threshold: tau_SCRC = 0.3175 (scrc_threshold.json)
5. Dual Safety Gate Action Rule: Accept if (u <= 0.3175 AND p_crop >= 0.80 AND p_disease >= 0.70) else REJECT

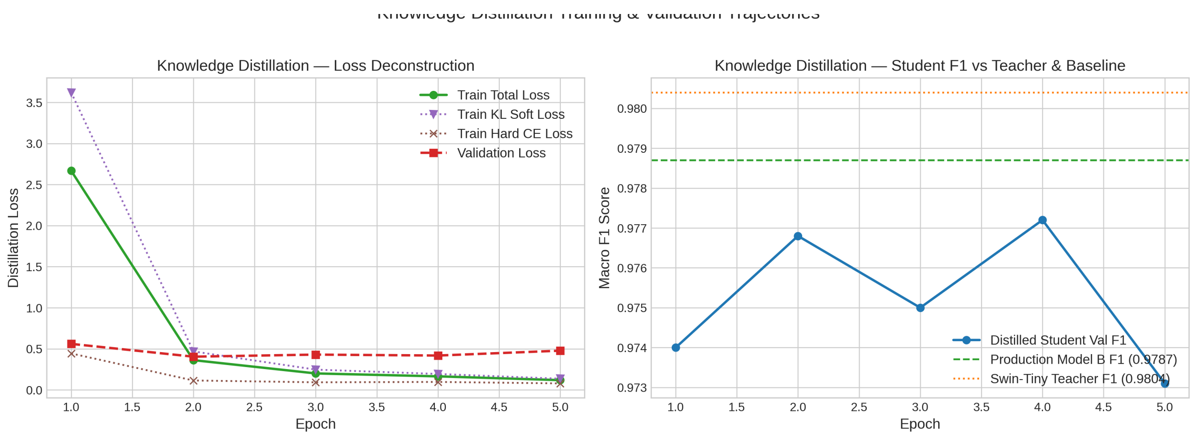

Figure: Knowledge Distillation Loss Trajectories & Evidential Calibration Curves

5. Multilingual Vector RAG & LLM Advisory Engine

Ground-truth Integrated Pest Management (IPM) advisories are retrieved from ChromaDB (ml_pipeline/rag/chroma_db/) using sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 (384d):

- Total Knowledge Chunks: 208 Chunks (26 Disease Records x 8 IPM Sections)
- Primary LLM: Meta Llama 3.1 8B Instant on Groq LPUs (~750 tok/s)
- Offline Fallback: Deterministic Evidence Synthesizer (100% Edge Availability, 0.86ms Execution Time)

6. Appendix: Defense Presentation Slides & Architecture Diagrams

Slide 1: System Workflow Architecture Block Diagram

Step #	Pipeline Module	Input / Tech	Parameter Count	Latency
Step 1	Client Upload	Next.js 14 / HTML5	0 Params	< 1.0 ms
Step 2	Tensor Preprocessing	PyTorch Torchvision	0 Params	0.45 ms
Step 3	Model A Crop Router	EfficientNetV2-B2	7,772,858 Params	1.25 ms
Step 4	Model B Classifier	EDLEfficientNetB2	7,786,948 Params	1.54 ms
Step 5	SCRC Risk Gate	Dirichlet Uncertainty	0 Params (tau=0.3175)	0.45 ms
Step 6	Vector RAG Search	MiniLM + ChromaDB	22,713,216 Params	5.32 ms
Step 7a	Offline Advisory	Deterministic Synthesis	0 Params	0.86 ms
Step 7b	Online Cloud LLM	Llama 3.1 8B (Groq)	8,030,261,248 Params	380.00 ms

Slide 2: Active Pass vs Total System Parameter Comparison

Execution Mode	Active Models	Active Parameter Count	Weights Memory	End-to-End Laten
Offline Edge Mode	Model A + Model B + MiniLM	38,273,022 Params	177.93 MB	9.40 ms
Online Cloud Mode	Model A + Model B + MiniLM + Llama	8,068,534,270 Params	Groq Cloud	388.56 ms